

Excel Mini Project – Step-by-Step Guide

Part 1: Data Cleaning

- Remove duplicate records based on PatientID using Data → Remove Duplicates.
- Convert VisitDate to 'dd-mmm-yyyy' format using Custom Format.
- Replace blank Diagnosis with 'Unknown' and blank Doctor with 'Unassigned' using Find and Replace.
- Delete the AGE RANGE column.
- Standardize text: Ensure Gender is only Male, Female, or Other. Use TRIM function to remove extra spaces in text fields.

Part 2: Formatting

- Bold headers and apply background color.
- Format Cost column as Currency with 2 decimal places.
- Format VisitDate as dd-mmm-yyyy.
- Apply conditional formatting: Highlight Age > 60 and Cost > 3000.

Part 3: Formulas (in a new sheet named 'Formulas')

- Age Group (Logical): =IF(Age<18, "Child", IF(Age<60, "Adult", "Senior"))
- Average, Max, Min Cost: Use AVERAGE(), MAX(), MIN().
- Count patients with Diabetes: =COUNTIF(DiagnosisRange, "Diabetes")
- Lookup Table and VLOOKUP for Standard Cost.
- Cost with Tax: =Cost * 1.1
- ROUND, ROUNDUP, ROUNDDOWN on Cost.
- Extract Year and Month: =YEAR(VisitDate), =TEXT(VisitDate, "mmmm")
- Rebuild Age Range using nested IF.

Part 4: PivotTables & Charts (in a sheet named 'Analysis')

- PivotTable - Total Cost by Doctor.
- PivotTable - Count of Patients by Gender.
- PivotTable - Average Cost by Diagnosis.
- PivotTable - Age Group Distribution.
- Insert PivotCharts: Pie Chart (Gender), Column Chart (Total Cost by Diagnosis).

Deliverables Checklist

- Raw Data sheet – untouched original.
- Cleaned Data sheet – cleaned and formatted.
- Formulas sheet – all calculations.
- Analysis sheet – pivot tables and charts.